Neural Network analysis of Thomson Scattering Spectra

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1 Abstract

Thomson Scattering (TS) is a powerful diagnostic for determining key plasma parameters, namely electron density and temperature. It is noninvasive and repeatable, and combined with high repetition-rate facilities, can be used to volumetrically map entire plasma regions. Here, we analyze Thomson scattering spectra of laser-driven plasma blast waves using neural networks (NN) to predict plasma density n_e and electron temperature T_e on test data. The objective of this project is to determine the utility of NN in extracting these key parameters and whether they can serve as a viable replacement for traditional manual fitting.

We evaluated three machine learning models. The first trained exclusively on experimental data, and the second entirely on calibrated synthetic spectra. The third was trained on synthetic spectra via transfer, where a model pretrained on synthetic data fine-tuned using experimental spectra. Model performance was evaluated using prediction error metric for electron density and electron temperature, and inference time was compared against traditional fitting methods and found to be significantly shorter.

We found that the model trained on purely experimental data was the most accurate in predicting Thomson spectra parameters. Nevertheless, the synthetic model predicted experimental spectra with comparable accuracy, demonstrating that training on synthetic spectra is a viable approach. Finally, it was found that transfer learning had the worst performance among the three models in terms of accuracy. All neural network models significantly reduced analysis time compared to manual spectral fitting, showing that machine learning methods can help to accelerate plasma diagnostics. The results of this project take into account machine learning considerations such as dataset differences between synthetic and experimental data, and the impact of training data selection on model performance.

2 The Big Picture

As high-power laser facilities move toward higher repetition rates, plasma experiments are producing data on a scale that older analysis workflows are not well equipped to handle. Rather than collecting only a small number of shots over long timescales, these more technologically advanced systems can generate large numbers of laser-target interactions in relatively short periods. This makes fast and reliable analysis increasingly in important for high-energy-density plasma research [1]. This is exactly what motivates the present project, which studies whether machine learning can accelerate the inversion of two-dimensional Thomson scattering spectra into plasma parameters.

Thomson scattering is a standard non-invasive diagnostic used to infer quantities such as the electron temperature T_e and electron density n_e from measured spectra [1]. In the conventional workflow, these parameters are recovered by fitting the theoretical spectral density function in Equation 1 to each measured spectrum [1, 2]. However, two distinct regimes, the collective and non-collective, make that challenging as they

have distinct spectral shapes. In what is known as the collective regime ($\alpha > 1$), the spectra exhibit stronger non-Gaussian structure, while in the non-collective regime ($\alpha < 1$), they are more nearly Gaussian: hence the relationship between spectral shape and plasma conditions changes across parameter space [1]. In the source paper, the broader goal was to improve the speed and accuracy of analysis for 2D Thomson scattering data, and our reproduction follows that same scientific motivation [1]. Framed as a machine-learning task, this is a supervised offline regression problem in which each spectrum is paired with fitted plasma-parameter labels and the model learns a mapping from spectral shape to physically meaningful quantities.

The prior work for this project is the standard PlasmaPy-based Thomson autofit procedure, which is physics-grounded but computationally expensive and often unstable for difficult spectra [1, 2]. The paper notes that some fits do not converge, especially for noisy spectra, spectra near the edges of the plane, and spectra contaminated by stray light leaking through the notch filter [1]. It also notes that density can be difficult to determine in the non-collective regime from spectral shape alone [1]. Eisenbach et al. proposed addressing this limitation with neural networks trained on experimental data, synthetic data, and transfer learning, while also situating their work within the larger machine-learning literature on Thomson-scattering-related tasks such as stray-light removal, noise modeling, and faster plasma diagnostics [1]. In practice, a successful model would not replace the underlying plasma physics or human judgment, instead it would serve as a fast first-pass or scalable analysis tool when repeated manual fitting becomes impractical.

Success in this work is defined both by prediction quality and by practical usefulness. Because T_e and n_e are continuous-valued targets, mean absolute error is a natural primary metric, and prediction plots are also important because they show where performance changes across plasma regimes. Speed also matters: the motivation for learning the inverse map is to reduce the repeated cost of manual or iterative fitting [1]. These measures are aligned with the scientific objective because the goal is not only to predict labels numerically, but to recover plasma conditions accurately enough to remain useful while substantially reducing analysis time. In a reproduction study, the minimum useful level of performance is therefore not exact agreement with every reported number in the source paper, but rather reasonable fidelity on the reserved set of real spectra together with a meaningful shortening of computational time. At the same time, several challenges complicate the problem: the physics differs between collective and non-collective spectra, stray light and noise can distort measurements, and synthetic spectra do not perfectly reproduce experimental conditions even when instrument effects are added [1]. This framing also assumes that the fitted labels used for training are reliable enough to act as ground truth and that performance on testing spectra is an indicator of how well the model would generalize in practice. For that reason, a reproduction is also judged by whether it recovers the same overall scientific pattern under different data and implementation constraints.

3 Data retrieval

The data from this project comes from the experiment detailed in [3], includes the spectra, measured wavelengths, n_e, and T_e in four separate .npy files, and is one-dimensional numerical array samples with associated plasma-parameter labels. The dataset is approximately 5.8 MB; small enough to store and work with easily in a standard Python workspace. This experimental dataset of 1,350 spectra was generated during a high-repetition laser experiment where and exploding laser plasma was produced by irradiating a solid polyethylene target with a high-energy (5–10 J) laser pulse at a high repetition rate of 1 Hz. The amount of data is set by the number of shots taken in the original experiment, with the set-up shown in 1.

From this existing experimental data, synthetic data was generated using the thomson.spectral_density() function in PlasmaPy to generate each synthetic spectrum, which were then modified by adding a notch filter and noise. The functions was set with a probe wavelength of 532 nm, a scattering angle of 90, and fixed ion temperature of 0.1 eV. The electron temperature and density input was from the experimental data set. The spectral density $S(k, \omega)$ describes the distribution of scattered light across the range of frequencies.

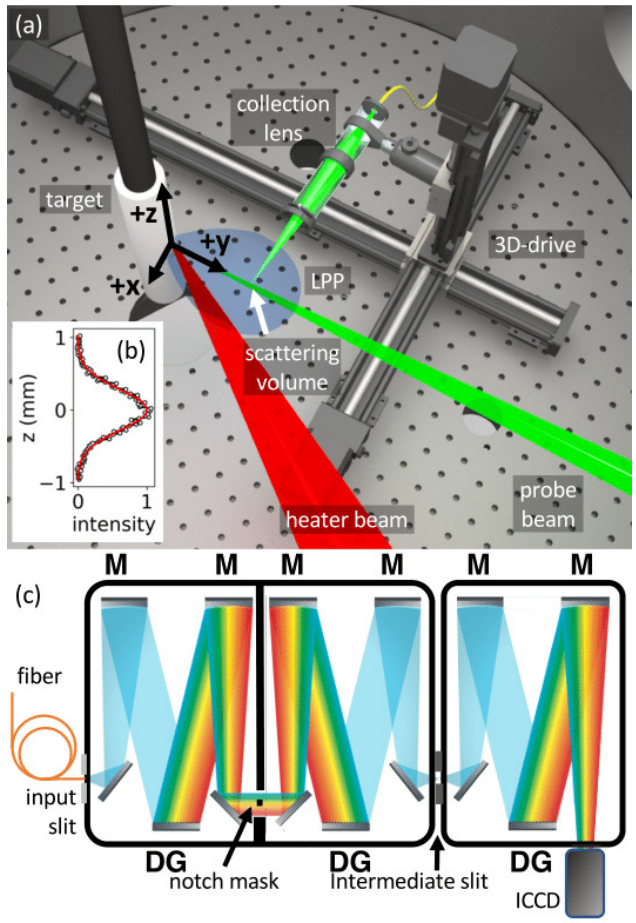


Figure 1: a) Schematic of the experimental setup b) demonstration of the translation of the Thomson probe along the $\hat{x}$-axis c) Demonstration of the light path with the notch mask. Reproduced from [4].

$$S(k,\omega) = \sum_e \frac{2\pi}{k} \left|1 - \frac{\chi_e}{\epsilon}\right|^2 f_{e0,e}\left(\frac{\omega}{k}\right) + \sum_i \frac{2\pi Z_i}{k} \left|\frac{\chi_e}{\epsilon}\right|^2 f_{i0,i}\left(\frac{\omega}{k}\right) \tag{1}$$

Where k is the scattering wavenumber, ω is the frequency difference between the incident and scattered radiations, and f_{e0}andf_{i0} are the equilibrium velocity distribution of electrons and ions, respectively, as seen in 3. To scale the spectral density by the electron density, they were multiplied together as $n_e \times S(k,\omega)$.

$$f_{e0}(v) = \frac{n_e}{v_{te}\sqrt{\pi}} \exp\left(-\frac{v^2}{v_{te}^2}\right) \tag{2}$$

$$f_{i0}(v) = \frac{n_i}{v_{ti}\sqrt{\pi}} \exp\left(-\frac{v^2}{v_{ti}^2}\right) \tag{3}$$

$$v_{te} = \sqrt{\frac{2k_B T_e}{m_e}}, \qquad v_{ti} = \sqrt{\frac{2k_B T_i}{m_i}} \tag{4}$$

Near the probe wavelength, the unscattered light from the interaction produces large counts, which can wash out the values out side this central range. However, the data in this area from the Thomson scattering

scattered light can be approximated as an inverted Gaussian [5]. An inverted Gaussian notch of FWHM 1.13 nm ($\sigma = 0.48nm$) was used to remove these large bins and replace them with an approximation of the two peaks on the side of the probe wavelength.

Finally, the synthetic spectra were convolved with an instrument function, which represents the finite resolving ability of the Intensified Charge-Coupled Device (ICCD) that was used to record the experimental data. Before model development, the experimental dataset was split into 75% training and 25% test data, and 10% of the training set was further set aside for validation during model training. Experimental spectra will include sharp peaks that are spread over multiple bins due to the finite slit width of the detector, so by convolving the synthetic spectral with the instrument function, the synthetic data better matches how experimental data will appear, which will contribute to higher accuracy.

4 Data exploration

Each spectra in the dataset consists of a 512-bin Thomson scattering spectrum together with fitted values of n_e and T_e through a traditional PlasmaPy method with no NN. We began by plotting the raw spectra to develop intuition for the signal behavior, and to identify erroneous features. We immediately observed overall noise, large spikes near 532 nm, and anomalous intensity spikes in select spectra. These informed areas to target during data cleaning. See Figure 2 for a visualization of these spectral features.

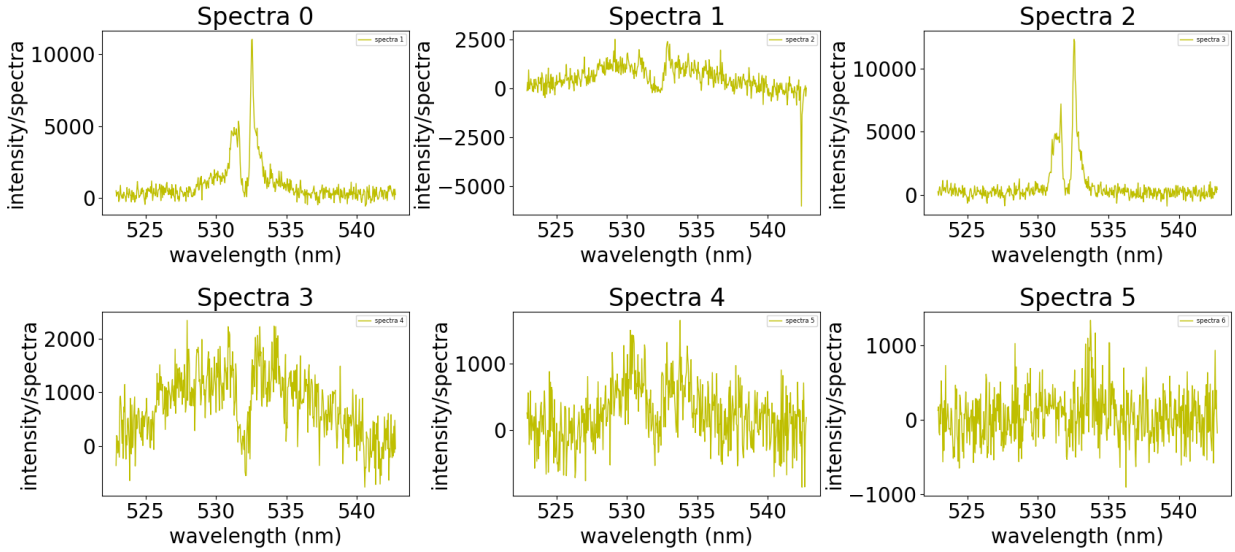


Figure 2: Overview of collected spectra showing noise characteristics, central stray-light features near 532 nm, and occasional anomalous intensity spikes.

By plotting electron density (in powers of 10) versus temperature, colored by α, we observed that the vast majority of spectra are in the non-collective regime. This raised concern about whether sufficient collective spectra existed for the network to learn from, given the richer structure in the spectral density function in that regime.[1] The distribution of electron density and temperature in Figure 3 therefore reveals an imbalance inherent in the dataset. We see that most spectra lie in the low-α non-collective regime, while comparatively

[1] All plasma in this experiment is assumed to follow a Maxwell-Boltzmann velocity distribution function, justified by the fact that the blast wave evolves on timescales much longer than the plasma thermalization time, so particles are well-approximated as being in local thermal equilibrium. Nonmaxwellian plasmas require new analysis techniques.

fewer samples fall in the higher-α collective regime. This raised concern about whether sufficient collective spectra existed for the network to learn from, given the more distinct structure in this regime.

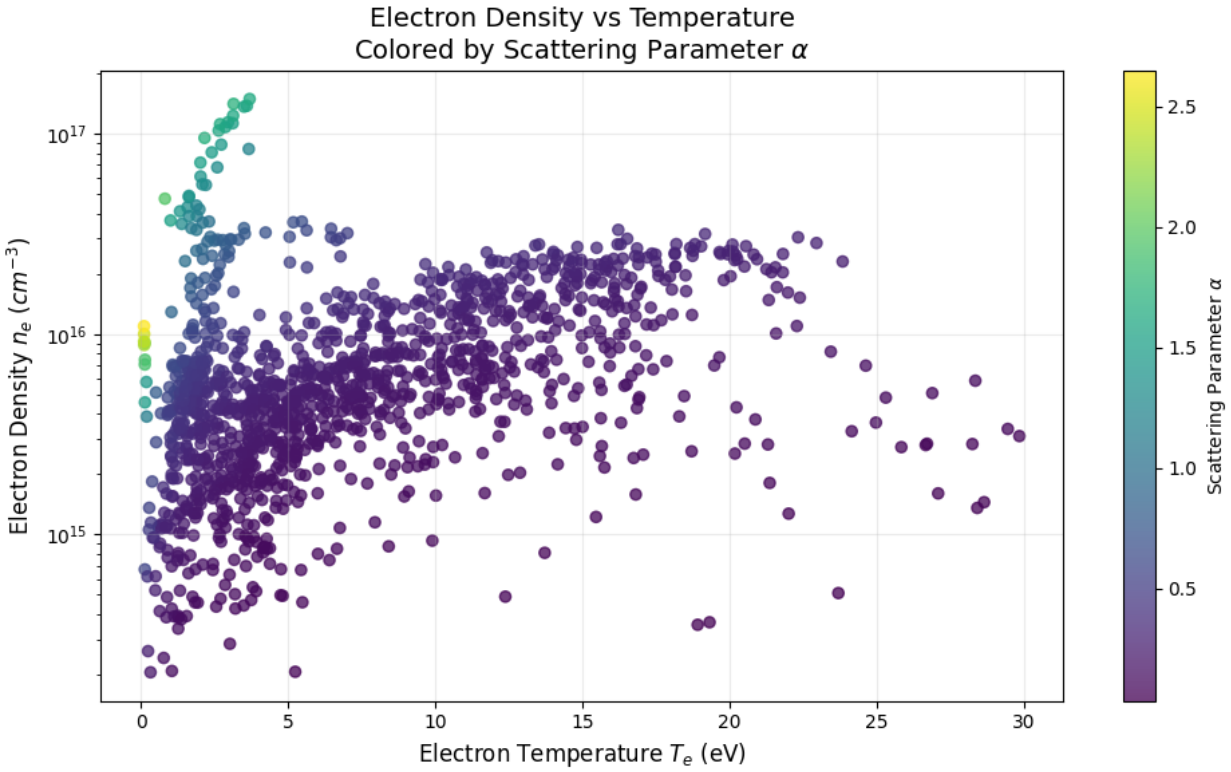


Figure 3: n_e versus T_e colored by α, showing that most spectra lie in the non-collective regime.

This imbalance is also reflected in the correlation relationships shown in Figure 4, Figure 5, and Figure 6. In this setting, the most informative patterns were not simple scalar correlations, but changes in overall spectral shape, width, and side-peak structure as the plasma conditions varied.

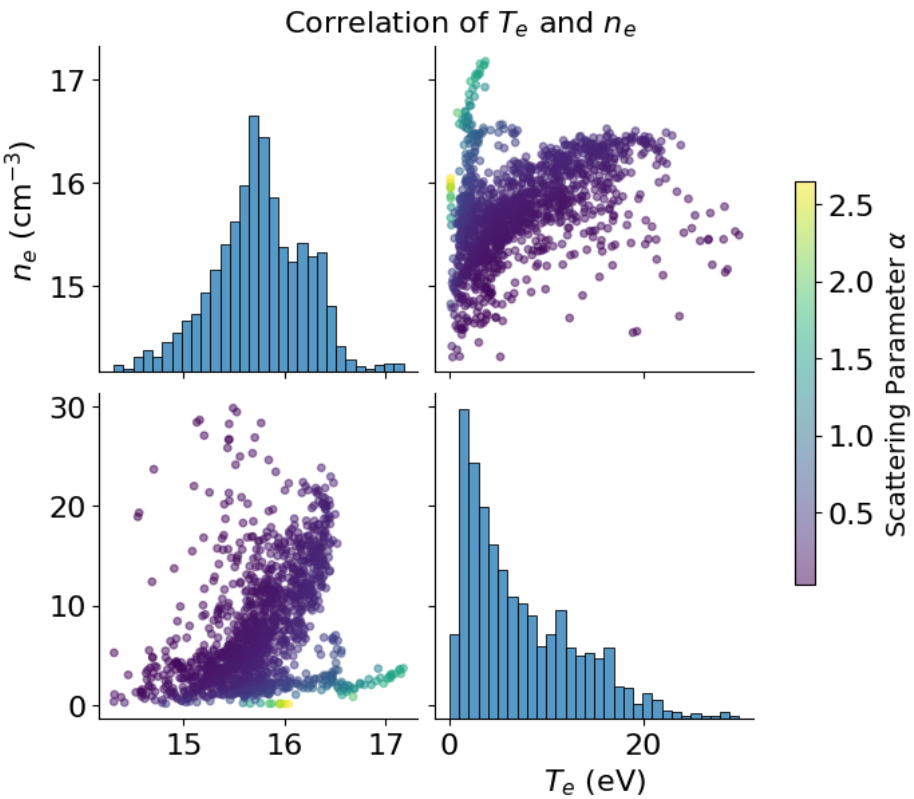


Figure 4: Pairwise relationships among the plasma parameters in the dataset.

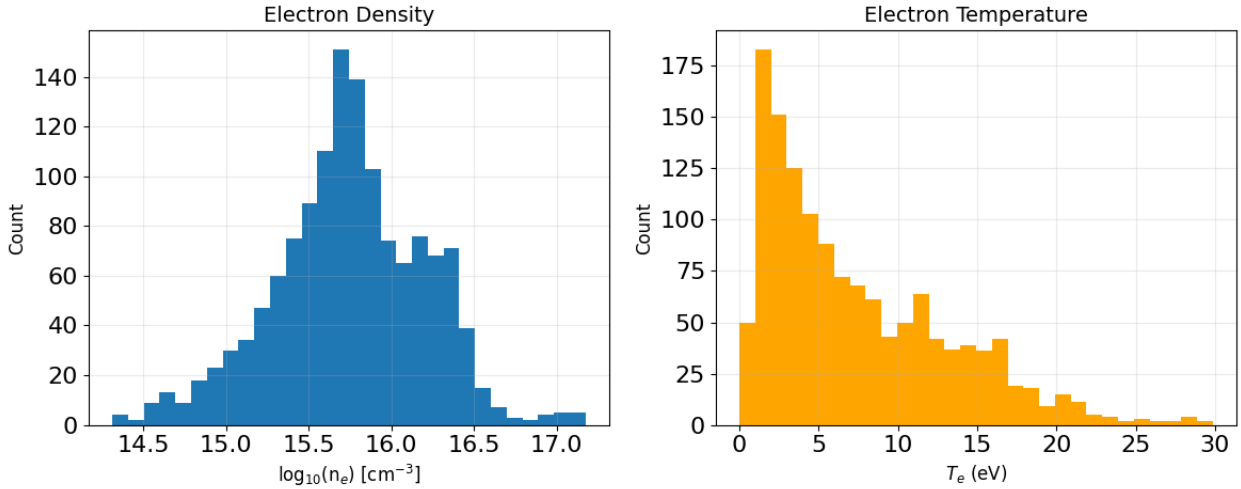


Figure 5: Distributions of electron temperature and electron density in the experimental dataset.

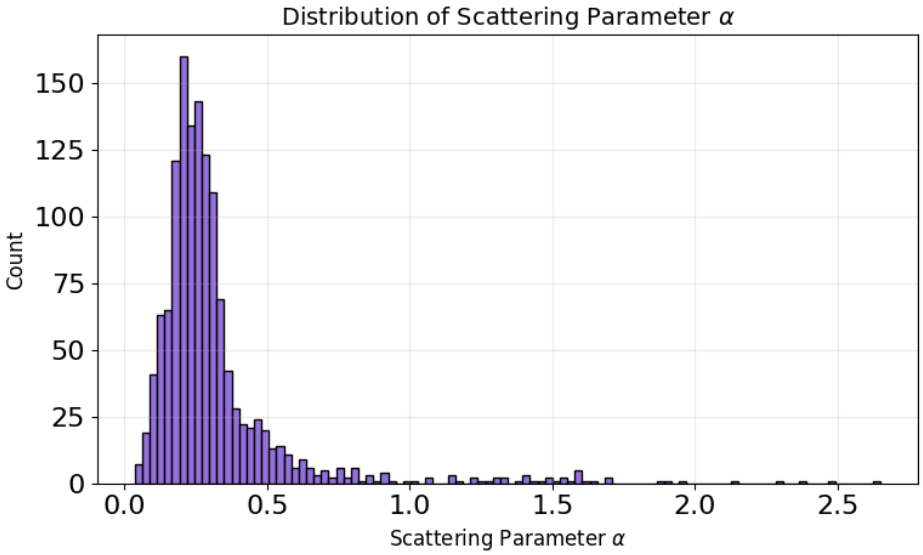


Figure 6: Distribution of the scattering parameter α, highlighting the dominance of the non-collective regime.

The distribution of electron density and temperature in Figure 3 shows that our dataset spans multiple Thomson scattering regimes, as indicated by the spread in the scattering parameter α. Here, we can see the true imbalance inherent in the dataset: most spectra lie in the α, non-collective regime, while comparatively fewer samples fall in the higher α, collective regime. This matters because collective spectra contain sharp EPW features which are more distinct features to fit, whereas non-collective spectra are broader and often less sensitive to small parameter changes. As a result, the dominance of non-collective cases may bias the models toward it with less confidence of collective spectra. This highlights the difficulty in modeling concentrated Thomson Scattering datasets with disproportionate regimes.

We also considered removing anomalous intensity spikes appearing in select spectra, as shown in Figure 2, Spectrum 1. However, it was decided to retain them, as they may encode information about the physical setup common to all spectra, and removing them could inadvertently discard meaningful signal. This reflects an important part of the exploratory stage: not every unusual feature should automatically be treated as bad data if it may contain physically relevant information.

Possible feature engineering was also considered. A natural option would have been to replace the full spectra with hand-chosen summary features such as peak widths, side-peak separations, or wavelength-band intensity ratios. However, because the goal of the project was to reproduce a neural-network approach that learns directly from spectral shape, the full 512-bin spectra were kept as the main input representation. In practice, the main feature engineering therefore came through preprocessing steps such as notch filtering, instrument convolution, and noise modeling rather than manual extraction of reduced spectral descriptors.

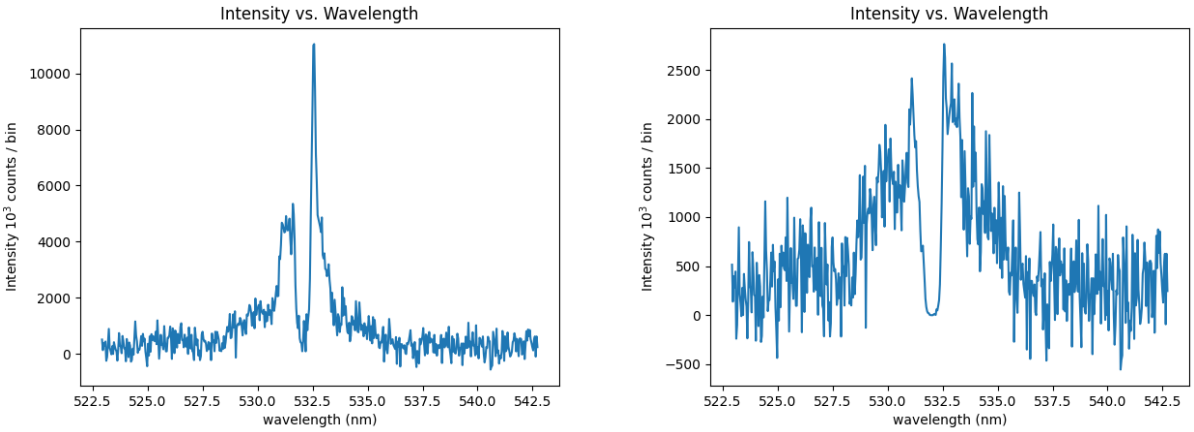
5 Data preparation

The primary data cleaning concern was the central spikes near 532 nm. All collected light passes through a triple grating spectrometer equipped with volume Bragg grating notch filters to suppress the intense probe laser signal at 532 nm, which would otherwise overwhelm the Thomson scattering signal. On some spectra, intense stray light leaked through at the filter edges, producing large spikes (see Figure 7a). To compensate, we multiplied the affected spectra by an inverted Gaussian to remove these spikes in proportion to the notch filter's strength within the range of its wavelengths (see Figure 7b). This specific method was motivated by

the advice of Dr. Niemann.

Preparation of the synthetic data focused on making the generated spectra more comparable to the experimental measurements. The synthetic spectra were convolved with an instrument function, which represents the finite resolving ability of the spectrometer used to create the spectra. Idealistically, each wavelength captured would be a sharp peak, but in practice it has spread over multiple bins due to the finite slit width of the detector, so by convolving the synthetic spectra with the instrument function, the synthetic data better matches how the experimental data appear. Additional instrument-inspired modifications, including a notch filter and added noise, were also applied so that the synthetic spectra more closely resembled the experimental inputs used during training.

Feature engineering in this project therefore entered mainly through preprocessing rather than through the construction of new hand-crafted descriptors. Rather than replacing each spectrum with summary quantities such as widths, peak separations, or intensity ratios, we preserved the full spectral arrays and used cleaning and transformation steps to emphasize the parts of the signal most relevant to plasma inference. This choice was consistent with the model architecture, since a fully connected neural network can learn directly from the full 512-bin input once the major artifacts have been reduced.



(a) Raw spectrum with stray light leakage near 532 nm.

(b) Spectrum after notch cleaning.

Figure 7: Comparison of a raw spectrum and the corresponding processed spectrum after notch cleaning.

6 Model selection and training

Our main modeling choices were made following the architecture and transfer-learning strategy presented in [1], since a central goal of the project was reproduction. Even so, we first trained a small Random Forest baseline in order to test whether a simpler non-neural model could recover the plasma parameters directly from the 512-bin spectra. Because the task is a supervised regression problem, the relevant comparison was between regressors rather than classifiers, and the baseline models were implemented with `RandomForestRegressor`. These models achieved reasonably strong R^2 values for n_e, but performed substantially worse for T_e. Also, the gap between train and test performance indicated overfitting. This hints at the fact that tree-based models could capture some coarse structure in the spectra, especially for density, but were less successful at learning the more subtle spectral patterns needed for temperature recovery.

Metric	Train	Test
R^2	0.97	0.80

Table 1: Random Forest results for n_e with max depth 100 and 100 estimators.

Metric	Train	Test
R^2	0.94	0.54

Table 2: Random Forest results for T_e with max depth 100 and 100 estimators.

We also tested a shallower Random Forest with a maximum depth of 10 and, in a third variation, reduced the number of estimators to 5 while keeping the depth fixed at 10. These changes slightly reduced the fit to the training data, but they did not remove the general pattern: the model remained noticeably stronger on n_e than on T_e, and the test performance for temperature stayed limited. In that sense, the Random Forest baseline was useful mainly as a reference point. It showed that the regression problem was not trivial, and that a model that treats the spectrum as a collection of split-based features struggles to generalize as well as desired, particularly for temperature.

Metric	Train	Test
R^2	0.86	0.75

Table 3: Random Forest results for n_e with max depth 10 and 100 estimators.

Metric	Train	Test
R^2	0.93	0.53

Table 4: Random Forest results for T_e with max depth 10 and 100 estimators.

Metric	Train	Test
R^2	0.86	0.68

Table 5: Random Forest results for n_e with max depth 10 and 5 estimators.

Metric	Train	Test
R^2	0.89	0.45

Table 6: Random Forest results for T_e with max depth 10 and 5 estimators.

Given the limitations of the Random Forest baseline, we then focused on neural networks, which were better matched both to the reproduction goal and to the transfer-learning idea emphasized in [1]. Transfer learning is especially useful when the data of primary interest are limited, since it uses a larger auxiliary dataset to learn useful structure before adapting to the true dataset of interest. In our case, this meant pretraining on synthetic Thomson scattering spectra and then retraining on experimental data.

Following [1], we trained three neural-network models: an experimental-only model, a synthetic-only model, and a transfer-learning model. The synthetic-only model was trained on PlasmaPy-generated spectra, the experimental-only model was trained directly on measured spectra, and the transfer-learning model combined the two by first training on synthetic data and then retraining on experimental data after freezing the first three learned layers. This gave us three distinct data approaches to compare: learning only from real data, learning only from synthetic data, and using synthetic pretraining to support learning on limited experimental data.

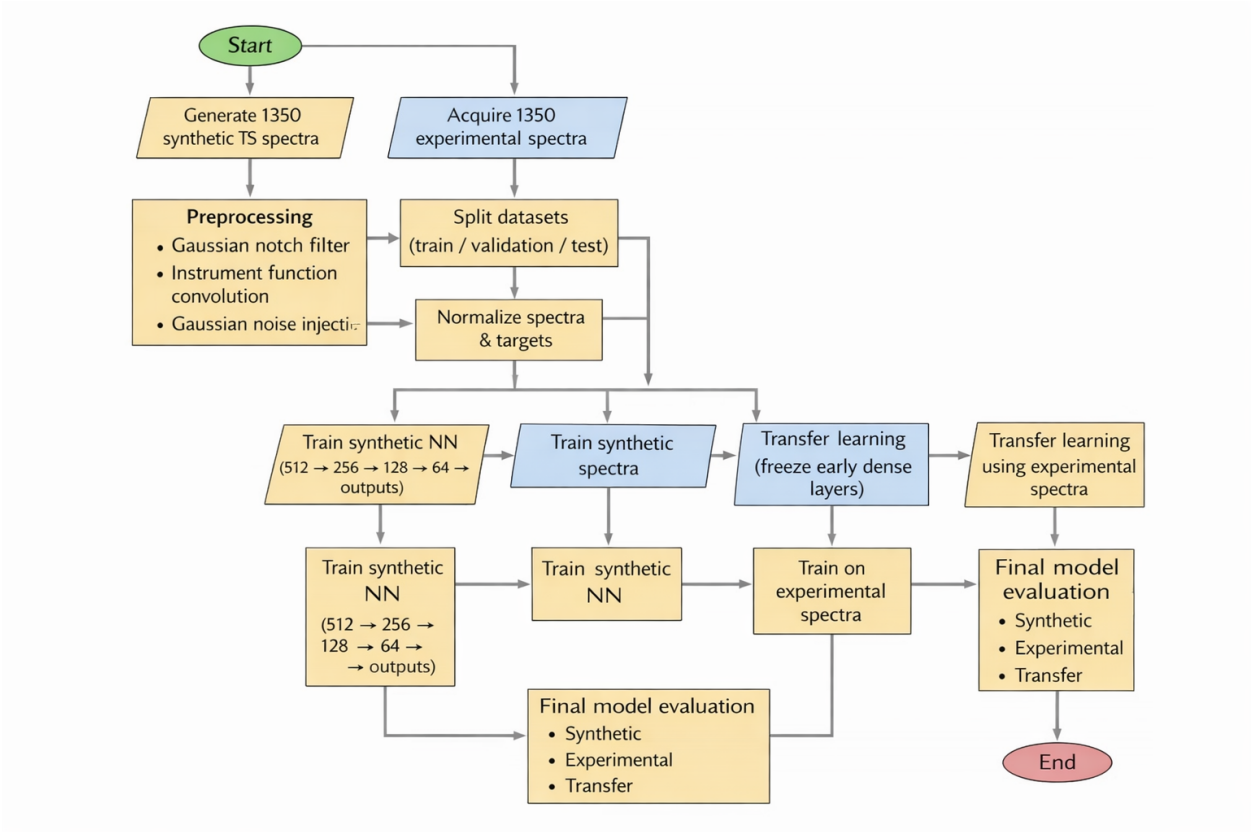


Figure 8: Flowchart showing the data processing and analysis for Transfer learning. TL model uses 1350 synthetic and experimental data generated from the same electron density and temperature values, with the model frozen layers taken from the synthetic model before training the remaining layers on the experimental data. Data was split into training and testing splits

Like the paper, our neural-network architecture consisted of an input layer of 512 nodes, one for each spectrum bin, followed by four dense hidden layers with ReLU activation. Each hidden layer then had half the number of nodes of the previous one. The output layer used a linear activation with four outputs: $\log_{10}(n_e)$, T_e, T_i, and α. Including T_i and α was made to keep the model structure aligned with the source paper and the physical parameters used in synthetic-spectrum generation. This architecture is shown in Figure 9 [1].

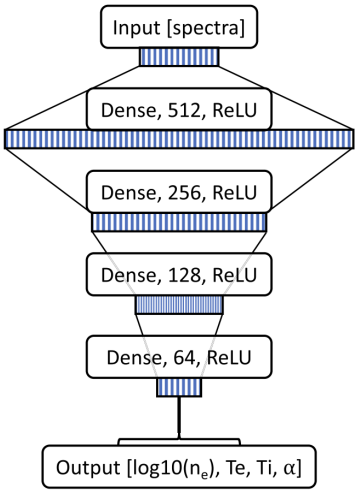


Figure 9: Neural-network architecture used for the experimental, synthetic, and transfer-learning models.

For evaluation, we used the test protocol defined before. Validation was performed during training on the training portion of the data. The experimental-only model was trained directly on the experimental training split. The synthetic-only model was trained on the synthetic dataset and then evaluated against experimental targets in order to test how well synthetic learning transferred to real data. The transfer-learning model used the same basic architecture, but after pretraining on synthetic spectra. Then the first three layers were frozen and the remaining layers were retrained on the experimental training data. Final performance for the neural-network comparison was then assessed on the experimental test set.

We used several metrics and visualizations to compare model quality. For the Random Forest baseline, we report train and test R^2 values for both n_e and T_e. For the neural-network models, we compared train accuracy, test accuracy, and mean absolute error (MAE), and we also inspected prediction plots to understand the types of errors each model made. Because this is fundamentally a regression problem, MAE was the more interpretable summary metric, while the prediction plots were important for revealing regime-dependent failures that a single scalar score could hide.

The neural-network comparison is summarized in Table 7. In our reproduction, the experimental-only model performed best overall, achieving the lowest MAE and the strongest test performance on real spectra. The synthetic-only model performed second best: it learned useful structure but did not match the experimental model once evaluated on real data. The transfer-learning model performed worst in our run, pointing at that synthetic pretraining did not bridge the gap as successfully as reported in [1]. This difference from the paper is itself informative, since it suggests that transfer learning in this setting is sensitive to details of the synthetic generation, noise model, instrument effects and train-test conditions rather than being automatically beneficial.

Model	**Train Accuracy**	**Test Accuracy**	**MAE**
Synthetic	89.32%	84.91%	0.040274
Experimental	88.92%	89.05%	0.037345
Transfer	79.23%	77.22%	0.053252

Table 7: Summary performance metrics for the three neural-network models.

Beyond the scalar metrics, the prediction plots showed that all three neural-network models handled density more consistently than temperature, and that errors increased in more difficult regions of parameter space, especially at higher temperatures. These qualitative behaviors help explain why the experimental-only

model worked best: it was trained directly on the same distribution on which it was tested, whereas the synthetic-only and transfer-learning models were more exposed to mismatch between simulated and measured spectra. Taken together, these results justified our decision to center the final comparison on the three neural-network approaches rather than on the Random Forest baseline.

7 Model fine-tuning

Ultimately, we decided to follow in the footsteps of the paper, so our hyperparameters consisted of 1000 epochs, a batch size of 128, and a learning rate of 10^-4. Each network had same structure of 512 nodes, 4 dense layers of neurons with ReLU activation and with size 512, 256, 128 and 64, and the output layer of 4 nodes. Our outputs were n_e and T_e.

Additionally, we used the same hyperparameters across all three networks: a learning rate of 10^{-4} , 1000 epochs with ADAM optimizer and MSE Loss function. We did switch the learning rate from 10^{-4} to 10^{-3} since it was originally really small and the model was sensitive to change, and this switch did improve our model a bit. However, in the end we chose to stick with 10^{-4} as our learning rate in order to remain consistent with the paper.

While this is the case, we decided to explore some optimizations to see if there were any improvements to make. Specifically, we ran a grid search of various learning rates, batch sizes, and epochs.

Grid Search Hyperparameters			
Batch Size	64	128	0.040274
Learning Rate	0.0001	0.001	0.01
Epochs	100		

Table 8: Chosen hyperparameters options for Grid Search to iterate through.

For both the synthetic and experimental spectra, a range of 30-50 epochs was found to be optimal, a learning rate of 0.001, and a batch size of 64 and 128 respectively. Because the transfer learning ends by training on experimental data, a batch size of 128 was choosen for it.

8 Results

The NN was found to be much faster than manual fitting using PlasmaPy, fitting with seocnds. For 5 spectra, manual fitting took about 20 minutes, but with NN the fitting was in the range of seconds., representing a significant reduction in computing time. The models were trained on the training set and evaluated on the held-out test set to assess their ability to generalize to unseen spectra. Model performance was quantified primarily using mean absolute error (MAE) between the predicted and true plasma parameters, along with prediction accuracy relative to the manually fitted reference values. These metrics allow direct comparison of model performance across the synthetic, experimental, and transfer learning approaches.

From Table 9 we see that the **experimental model** achieved the best performance in terms of test accuracy and the MAE. The diagnostic plots shown in Figure 10, Figure 11, and Figure 12 compare the predicted plasma parameters with their true values in the dataset. All values of alpha were clipped to a maximum of 1 along the color-bar to highlight the transition from the non-collective to collective regime (or lack thereof). In each plot, the **identity line** ($y = x$) represents perfect agreement between the model predictions and the ground truth. The synthetic model converged faster, and yielded similar results as the experimental dataset,

with only a slightly higher MAE, suggesting that both datasets were likely similar in their representativeness of the physical experiment. Our experimental dataset had an extreme disproportionality in its representation of different plasma regimes as seen in Figure 3, which likely produced the significant deviations for high-density low-temperature values corresponding to the collective regime. There are several reasons for the deviations in the synthetic only model as well, such as the inherent noise added to the spectra that matched the process of [1]. In that same paper, they mention that when the synthetic model was trained without noise, the points were clustered more tightly around the identity line, but we keep the added noise as it matches the experimental spectra better. They also note that adding gaussian noise with a constant width to emulate experimental conditions may have affected the network's ability to determine the temperature explaining the significant deviations in the temperature diagnostic plots.

At low electron densities and high electron temperatures (corresponding to low values of the scattering parameter α, i.e., the non-collective regime), the model predictions deviate more significantly from the identity line. This indicates increased difficulty in accurately predicting plasma parameters in regimes where the Thomson scattering spectrum exhibits weaker collective structure. Conversely, the models perform substantially better for higher electron densities and lower temperatures (larger α), suggesting improved predictive performance in regimes where collective spectral features are more pronounced.

Unlike the results reported in [1], our transfer learning model produced the lowest predictive accuracy on both the training and test sets and exhibited the largest MAE. This discrepancy likely arises from several factors. First, the experimental dataset used in our analysis differs from that used in the referenced work, both in spectral composition and sample size. While the study in [1] utilized 1664 spectra, our dataset contained only 1,350 experimental spectra. Second, the synthetic training dataset in the referenced work consisted of 10,000 spectra generated from randomly sampled plasma parameters using `np.random`, whereas our synthetic dataset consisted of only 1350 spectra generated from a limited set of (n_e, T_e) values.

Although both studies applied similar preprocessing steps, including modeling the instrument response via Gaussian convolution, suppressing stray light using a Gaussian notch filter, and adding Gaussian noise to emulate measurement uncertainty, the differences in dataset size and parameter sampling likely contributed to the discrepancy in model performance. These results emphasize the importance of large, diverse, representative synthetic datasets when training neural networks to infer plasma parameters across different physical regimes.

Model	Train Accuracy	Test Accuracy	MAE
Synthetic	89.32%	84.91%	0.040274
Experimental	88.92%	89.05%	0.037345
Transfer	79.23%	77.22%	0.053252

Table 9: Performance comparison of neural network models trained on synthetic, experimental, and transfer datasets.

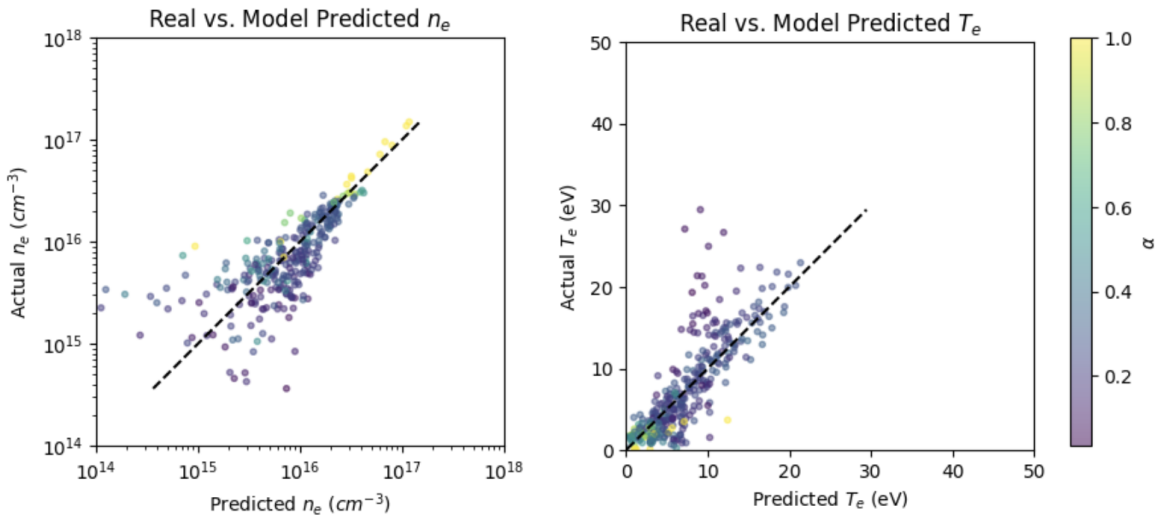


Figure 10: Synthetic NN Diagnostic Plot. The points are color coded by the plasma parameter α

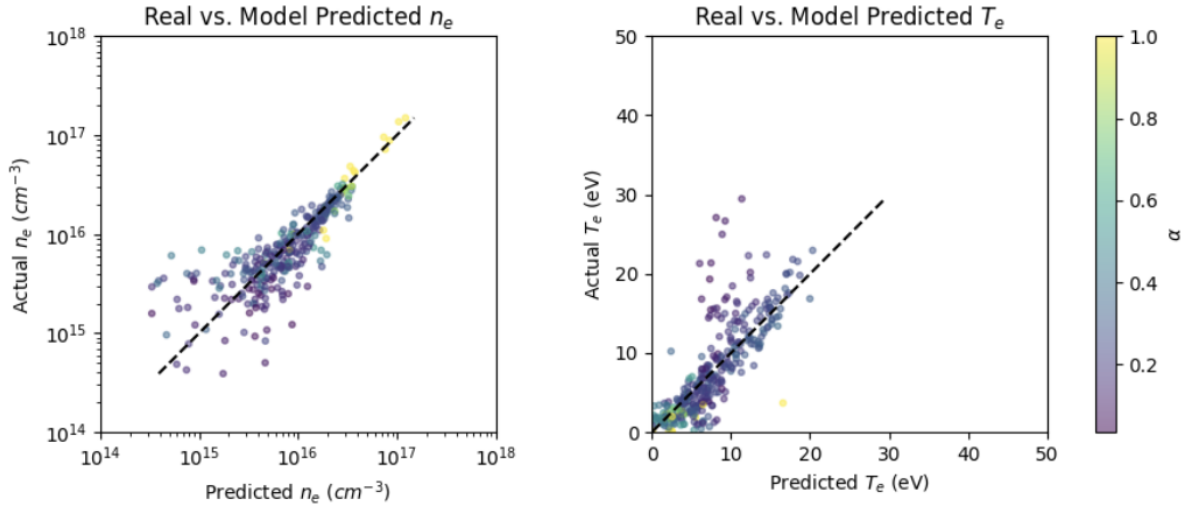


Figure 11: Experimental NN Diagnostic Plot. The points are color coded by the plasma parameter α

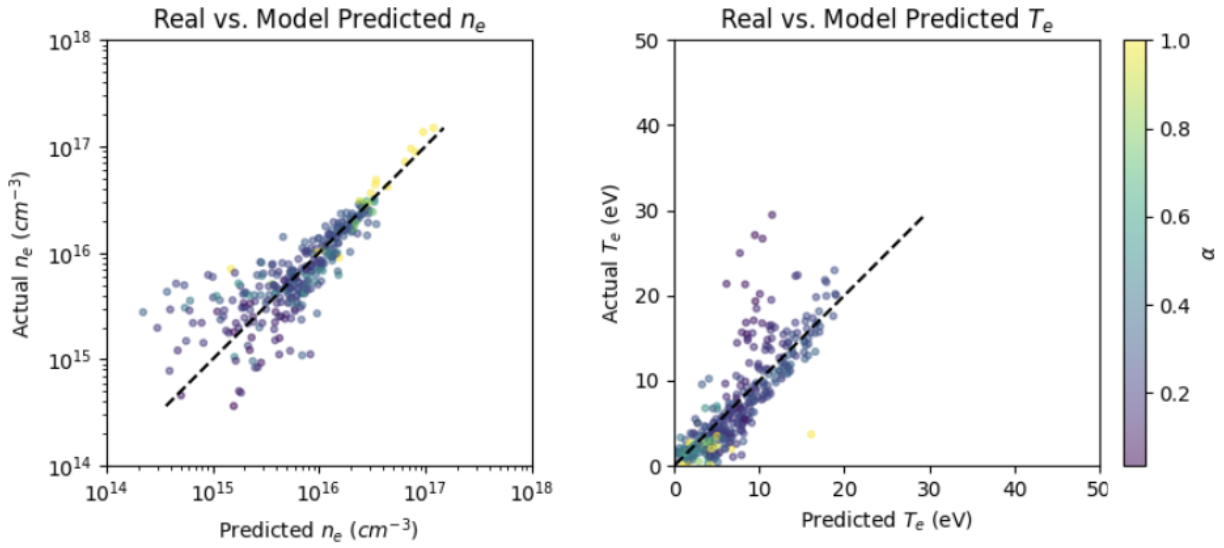


Figure 12: Transfer Learning NN Diagnostic Plot after retraining on experimental data. The points are color coded by the plasma parameter α

While our model performance did not directly benefit from transfer learning, our general machine learning approach still achieved the primary scientific objective of this experiment: enabling rapid extraction of electron density and temperature from Thomson scattering spectra. Unlike traditional nonlinear least-squares spectral fitting, which can require repeated expensive serial optimization and unstable initial parameter guesses, the trained neural networks produce predictions in a fraction of a second. This capability has the potential to significantly accelerate plasma diagnostics in experimental settings, enabling near real-time analysis of plasma conditions and improving the efficiency of experimental campaigns. Accurate estimates of electron density and temperature is fundamental for characterizing plasma state variables such as pressure, energy density, and collisionality, and for constraining key plasma parameters including Debye length and plasma frequency. Rapid inference of these quantities therefore enables improved monitoring of plasma evolution, validation of theoretical models, and more efficient control of laboratory plasma experiments.

There are several weaknesses with the transfer model approach that could be improved upon to increase model accuracy. First, the synthetic dataset did not appear to contribute significantly to model performance and in some cases appeared to hinder it. One potential improvement would be the generation of a substantially larger synthetic training dataset. In many synthetic-to-real transfer learning applications, large synthetic datasets are used to pre-train models because they provide perfectly labeled data that can be generated at scale with minimal cost [6]. Moreover, their synthetic results serve as a theoretical baseline based on expected physical results that strengthen the models foundational understanding contextually. Increasing the number and diversity of synthetic spectra would allow the neural network to learn a broader mapping between spectral features and plasma parameters before being fine-tuned on the smaller experimental dataset. Previous studies of synthetic-to-real transfer learning have shown that increasing the size of the synthetic pre-training dataset can improve model generalization and reduce prediction error when applied to real data, particularly when real labeled data are limited [7]. Implementing a similar approach to these models would likely improve model performance, not only for moderate moderate plasma conditions but also for edge cases corresponding to different physical regimes present in Thomson scattering experiments.

Nevertheless, the machine learning models developed in this approach rely on several assumptions that may not completely imply, and may therefore limit the predictive accuracy of these models in certain regimes. Outside of the collective vs. non-collective regimes mentioned before, the synthetic spectra used for training

both the synthetic and transfer (first three frozen layers) models assume idealized plasma conditions and rely on a theoretical Thomson scattering model (see Equation 1) that neglect additional experimental complexities such as background emission, imperfect instrument calibration (assumed gaussian instrument error), and spatial plasma gradients. Moreover, the models assume that the training dataset is adequately representative of the relevant plasma parameter space. If regions of electron densities and temperatures are underrepresented in the training data, the neural network may perform poorly when predicting spectra that fall near or outside boundaries of the sampled parameter space. This limitation is particularly relevant for edge cases which correspond to extreme but real plasma conditions. Neural Networks also operate as empirical function approximates and don't explicitly enforce physical constraints. Alternate approaches such as Physics-Instructed Neural Networks (PINN's) could prove beneficial for constraining the conditions of edge cases aforementioned. While the models can learn correlations between spectral features and plasma parameters, they do not guarantee physically consistent predictions outside the range of the training data. Consequently, the reliability of the inferred plasma parameters depends strongly on the representativeness and quality of the training dataset.

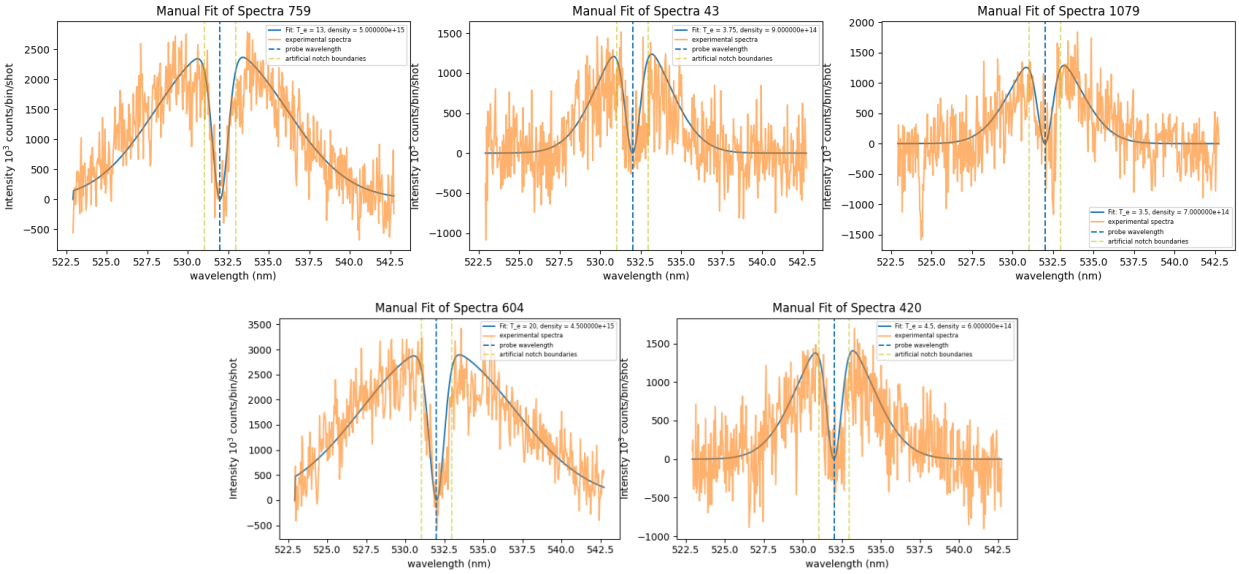


Figure 13: Manual fits across the validation spectra used as a baseline comparison, T_e in units of eV, n_e in units of density.

9 Contributions

Isaiah, Joey, and Kristiano focused on processing the data and preparing datasets, coding and developing the different models, training and running the models, adjusting hyperparameters and interpreting results. Clarissa and Setareh focused on retrieving data, doing background research, creating visuals, and writing up the final paper. We all worked together on the slides presentation and editing the final paper.

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